

INDUSTRY INTERNSHIP REPORT / THESIS

Submitted in partial fulfillment of requirement for the award of degree of

Bachelor of Technology in Computer Science and Engineering

by

Punyaa Dixit

Industry / Organization Guide

Dr. / Mr. / Mrs. _____

at

Local Large Language Model Hallucination Detection Project

Institute Guide (from College)

Dr. Apeksha Sakhare

Assistant / Associate Professor, Department of CSE

May 2026

Department of Computer Science and Engineering

G H Raison College of Engineering, Nagpur

An Empowered Autonomous Institute affiliated to RTMNU, Nagpur

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Declaration

I hereby declare that this Industry Internship Report / Thesis entitled ”**A Multi-Stage Framework for Detecting Factual Hallucinations in Large Language Model Outputs**” submitted in partial fulfillment of the requirements for the award of Bachelor of Technology in Computer Science and Engineering has been carried out by me.

The work presented in this report is original and has not been submitted, in whole or in part, for the award of any other degree or diploma at this or any other institution or university.

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Certificate

This is to certify that the Industry Internship Report / Thesis entitled ”**A Multi-Stage Framework for Detecting Factual Hallucinations in Large Language Model Outputs**” has been carried out under our supervision and guidance by **Punyaa Dixit** for the award of Degree of Bachelor of Technology in Computer Science and Engineering.

The work submitted is comprehensive, complete, and fit for evaluation.

Dr. / Mr. / Mrs. Dr. Apeksha Sakhare

Industry / Organization Guide

Designation:

Institute Guide

Assistant / Associate Professor, CSE

Organization:

GHRCE, Nagpur

Note: Attach the official industry certificate(s) on organization letterhead with reference/document number, internship duration, project details, and stipend/sponsorship details, as required by the departmental guideline.

Acknowledgement

I express my sincere gratitude to my institute guide, **Dr. Apeksha Sakhare**, for continuous guidance, constructive feedback, and support throughout this work. I also thank the Department of Computer Science and Engineering, G H Raison College of Engineering, Nagpur, for providing an encouraging academic environment.

I acknowledge the open-source community and research ecosystem, including Hugging Face, spaCy, Sentence-Transformers, and Wikipedia contributors, whose tools and knowledge sources were instrumental in developing and evaluating this hallucination detection framework.

Abstract

Large Language Models (LLMs) are widely adopted in content generation and decision-support workflows, but they often produce hallucinations: fluent statements that are factually incorrect or unverifiable. This thesis presents a multi-stage framework for detecting factual hallucinations in LLM outputs. The framework includes claim extraction, evidence retrieval, verification using natural language inference, and a risk-based hallucination scoring layer.

A modular Python implementation was developed using spaCy, transformer-based inference models, and Wikipedia-backed retrieval. Evaluation on curated scientific, historical, geographic, and intentionally hallucinated samples demonstrated strong performance, with **82.1% overall claim-level accuracy** and **87.3% precision for contradicted claims**. The system is designed for practical deployment and includes extensible interfaces for local and service-based model backends. This work contributes a practical, explainable verification pipeline that can improve trust and safety in real-world LLM applications.

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List of Symbols

T	Input LLM-generated text
C	Set of extracted factual claims
E	Evidence set retrieved for a claim
$V(c)$	Verification label for claim c
H	Overall hallucination score for a document
θ	Support threshold in verification aggregation

List of Publications / Competitions / Patents

- Research paper: *A Multi-Stage Framework for Detecting Factual Hallucinations in Large Language Model Outputs.*
- Add published paper details / competition participation / patent filing details here, if applicable.

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The chapter sequence of this report follows the institutional industry report guideline.

Chapter 1

Introduction to Company / Organization Context

This thesis was executed as an industry-aligned applied AI project in the domain of **trustworthy large language model systems**. The project context is software-product engineering, where generated content must be validated before deployment in user-facing workflows.

The practical problem addressed is: *how to detect factual hallucinations in generated text without retraining the base LLM*. The developed solution is a modular post-hoc verification pipeline suitable for integration with local and cloud model ecosystems.

1.1 About the Project Environment

The implementation environment consisted of Python 3.12, model inference libraries, NLP toolkits, and a lightweight web interface for interactive analysis. The workflow included data preparation, pipeline experimentation, error analysis, and reproducible reporting.

1.2 Historical Background

As LLM capabilities expanded rapidly between 2022 and 2026, concerns about hallucinated content also increased. Prior studies showed that fluency does not imply truthfulness. This motivated development of independent verification systems that can audit generated claims against trusted evidence.

1.3 Location and Operational Structure

The project was carried out within the academic and research setup of G H Rasoni College of Engineering, Nagpur, with guidance from faculty mentorship. Operationally,

work followed iterative phases: literature review, architecture design, implementation, evaluation, and final documentation.

1.4 Vision and Mission Alignment

Vision: Improve reliability and safety of AI-generated content through transparent verification.

Mission:

- Build a practical framework for detecting hallucinations in LLM outputs.
- Provide interpretable, claim-level evidence for user trust.
- Support deployment in educational, research, and industry assistant applications.

1.5 Product / Solution Developed

The resulting product is a configurable hallucination detector with these major modules: claim extraction, evidence retrieval, NLI-based verification, and risk scoring. A web interface and script-based APIs are provided for integration and testing.

Chapter 2

Case Study / Project Work

2.1 Introduction

The project investigates factual hallucination detection as a structured verification problem. Instead of modifying language model generation, the system evaluates completed model outputs and quantifies factual risk.

2.2 Problem Identification

LLM outputs can include fabricated entities, wrong dates, incorrect relationships, and unverifiable claims. In high-impact domains, such errors reduce trust and can cause downstream operational risk.

2.3 Objectives

- Extract atomic factual claims from free-form generated text.
- Retrieve high-relevance supporting evidence from trusted sources.
- Classify each claim as supported, contradicted, or unverifiable.
- Produce an overall risk score and interpretable report.
- Validate effectiveness across mixed-domain evaluation data.

2.4 Work Carried Out

2.4.1 Pipeline Design

A four-stage architecture was designed:

1. Claim extraction using linguistic and entity-aware methods.
2. Evidence retrieval using query-focused lookup and semantic filtering.
3. Verification via transformer-based NLI inference.
4. Aggregated hallucination scoring and risk labeling.

2.4.2 Hardware and Software Setup

- Hardware: Standard development workstation; optional GPU acceleration.
- Software: Python 3.12, Transformers, Sentence-Transformers, spaCy, NLTK, Wikipedia API.
- Interface: Streamlit-based web front end for interactive claim analysis.

2.4.3 Solution Implementation

The detector was implemented as modular classes in Python for maintainability: *ClaimExtractor*, *EvidenceRetriever*, *InferenceModel*, and *HallucinationScorer*. This design supports testing and replacement of individual components.

2.5 Solution Provided

The final solution is an end-to-end hallucination assessment engine that accepts generated text and returns:

- Extracted factual claims.
- Matched evidence snippets and relevance confidence.
- Claim-level verification labels (Supported/Contradicted/Unverifiable).
- Document-level hallucination score and risk band.

2.6 Calculation Done

Claim-level risk scores are mapped as:

$$\text{conf}(v_i) = \begin{cases} 0.0, & v_i = \text{SUPPORTED} \\ 0.5, & v_i = \text{UNVERIFIABLE} \\ 1.0, & v_i = \text{CONTRADICTED} \end{cases} \quad (2.1)$$

Overall hallucination score:

$$H = \frac{1}{n} \sum_{i=1}^n \text{conf}(v_i) \quad (2.2)$$

where n is the number of extracted claims.

Relevance scoring between claim and evidence embedding vectors:

$$\text{relevance}(c, e) = \frac{\text{emb}(c) \cdot \text{emb}(e)}{\|\text{emb}(c)\| \|\text{emb}(e)\|} \quad (2.3)$$

2.7 Results and Conclusion

Evaluation outcomes from the final selected research paper are summarized below.

Table 2.1: Verification Performance Summary

Status	Precision	Recall	F1
Supported	81.4%	78.6%	80.0%
Contradicted	87.3%	83.2%	85.2%
Unverifiable	74.8%	79.3%	77.0%
Overall	82.1%	80.4%	81.2%

Table 2.2: Accuracy by Content Domain

Domain	Claim Accuracy	Document Accuracy
Scientific Facts	86.3%	92.0%
Historical Events	79.2%	86.0%
Geographic Information	82.7%	88.0%
Hallucinated Content	81.4%	88.0%

The framework achieved reliable detection of contradicted claims and practical end-to-end throughput, showing suitability for real-world AI verification workflows.

Chapter 3

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Chapter 4

Appendices

Appendix A: Image/Photo List of Project Work

- Add screenshots of system architecture, web interface, and result dashboards.
- Add group photo with guide / co-guide (if applicable).

Appendix B: Industry/Organization Engagement Evidence

- Add photos from organization visits / interactions with supervisors.
- Attach signed industry certificates with reference/document number.
- Add cost-savings or implementation impact certificate, if available.

Appendix C: Publications / Competitions / Innovation Artifacts

- Add publication acceptance/email proof (if available).
- Add project competition participation/winning certificates.
- Add patent filing copy (if filed).